A3-PS035-TC476

Evidence-Based Operator Network Fault Localisation

The Edge Innovators

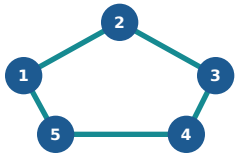
Shresth Rungta
Department of Mathematics and Computing

Alishma Islam Siddiqui
Department of Computer Science and Engineering

Mentor: Dr. Swati Kumari

AIORI Hackathon 3 • October 14, 2026

rungtashresth216@gmail.com



RULE-BASED BASELINE FIRST

EXPLAIN EVIDENCE • ADMIT UNCERTAINTY

Abstract & Problem Statement

Evidence-based diagnosis of network faults

Network faults can disrupt services and impact users, but identifying the exact cause is often difficult. Different faults can produce similar symptoms, making it challenging to locate the faulty router or link using a single observation.

THE CHALLENGE

Correlated interface alarms, BGP session drops, reachability failures and latency warnings can obscure the initiating fault.

OUR OBJECTIVE

Combine topology, syslog events and active ping measurements to rank plausible root causes and explain each prediction.

SAFETY BOUNDARY

No automatic remediation. Never claim a root cause can always be identified; report ambiguity or insufficient evidence.

REQUIRED SCOPE

- Five-node instrumented network with five seeded fault scenarios.
- Rule-based diagnostic engine before any optional AI model.
- Top-1 and top-3 accuracy on held-out runs, plus analysis of unsupported conclusions.

Source: AIORI Hackathon 3, PS-035 (Cybersecurity). [1]

Background

Why topology and telemetry must be read together

TOPOLOGY

Nodes, links and paths provide structural context for interpreting where symptoms appear.

SYSLOG / CONTROL PLANE

Interface state transitions, BGP neighbor changes and device messages provide event timing and control-plane clues.

ACTIVE PROBES

Ping success, packet loss and round-trip-time (RTT) trends across selected hops.

DIAGNOSTIC PRINCIPLES

LINK FAILURE

A link-down event plus failed probes across that same link supports a link-failure hypothesis.

CONTROL-PLANE FAULT

BGP session failure while interface-level reachability remains healthy points toward a control-plane fault.

DEGRADATION / UNCERTAINTY

Partial loss or elevated RTT may overlap with other causes; missing telemetry reduces confidence.

Telemetry framing follows RFC 9232: collect, correlate and consume data from multiple network viewpoints. [2]

Literature Review

Grounding the design in existing work

RFC 9232 — NETWORK TELEMETRY FRAMEWORK

Describes generating, collecting, correlating and consuming telemetry across management, control, forwarding and external-data viewpoints. Motivates a modular collection and analysis pipeline. [2]

IETF 126 HACKATHON — AI-ASSISTED FAULT LOCALISATION

Related work describes correlating symptoms, telemetry, topology, configuration, logs and alarms to identify probable causes and supporting evidence. Avoid claiming broad novelty for correlation alone. [3]

WHAT THIS PROJECT WILL TEST

Whether a small, transparent rule-based engine can distinguish five seeded fault families in a controlled five-node topology, provide auditable evidence for its rankings, and abstain when observations are insufficient.

This is a scoped engineering contribution, not a claim of state-of-the-art performance.

Proposed Solution

A transparent diagnosis pipeline

1 DEFINE TOPOLOGY

Five nodes in a ring; stable node/link IDs.

2 COLLECT OBSERVATIONS

Timestamped events, ping success, loss and RTT.

3 NORMALIZE & CORRELATE

Map observations to nodes/links; align time windows.

4 GENERATE HYPOTHESES

Compare evidence against candidate fault signatures.

5 RANK + EXPLAIN

Top-1/top-3 with supporting, contradicting and missing evidence.

6 EVALUATE SEPARATELY

Held-out labels; inspect errors, ties and abstentions.

Illustrative output (not a measured result): “BGP session failure — supported by a BGP-down event while interface probes remain reachable; link-down hypothesis contradicted by healthy link state.”

Solution Optimization

Make each diagnosis auditable

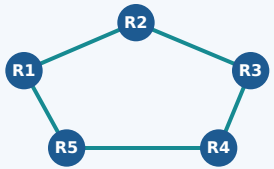
DIMENSION	CONVENTIONAL / MANUAL TRIAGE	PROPOSED APPROACH
Evidence handling	Operator cross-references multiple alarms and tools manually.	Normalize topology, syslog and ping observations into one evidence set.
Root-cause output	Alarm list may contain secondary symptoms and duplicate alerts.	Rank plausible root causes; show evidence for each candidate.
Ambiguity	May encourage a premature single-cause conclusion.	Expose ties, missing telemetry and insufficient evidence; allow abstention.
Validation	Operator judgement is difficult to compare consistently.	Held-out runs; top-1/top-3 accuracy; error and uncertainty analysis.

Validation guardrail: a score is not a probability unless calibrated and validated. Tune thresholds on development runs, then freeze them before held-out evaluation.

Solution Architecture

The diagnostic engine sees observations—not the hidden answer

1 NETWORK & SIGNALS



Topology + syslog/BGP + active ping/RTT. Routing state is optional lab input.

2 NORMALIZE & CORRELATE

Timestamp • node • link • symptom

RULE-BASED BASELINE

Candidate hypotheses + evidence

RANK + UNCERTAINTY

Top-1/top-3 • abstain if weak

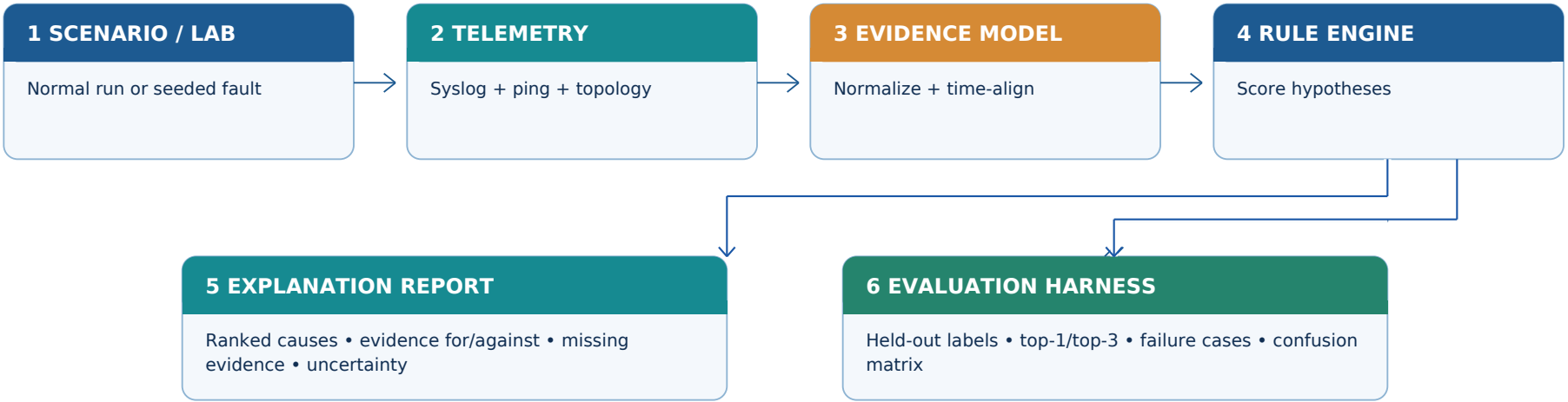
3 EVALUATION ONLY

Ground truth + held-out runs. Compare predictions with hidden labels; report top-1/top-3 and failure cases.

Ground truth must never be passed to the diagnostic engine. Python simulator + rule engine first; Containerlab/FRR can be added if time and environment permit.

Data Flow Diagram

From fault injection to an auditable report



CORE RECORDS

Topology nodes/links; timestamped event records; probe target, time, success/loss and RTT; diagnosis candidates with evidence lists; separate ground-truth labels for evaluation only.

LABELS ARE EVALUATION-ONLY — DIAGNOSTIC INPUTS EXCLUDE GROUND TRUTH.

Milestones, Validation and Risks

Complete proposal materials first; use the development window to harden the prototype

OCT 14 • LEVEL 1

Presentation, team-introduction video and repository collaboration steps. Exit: PDF/video link present; repository requirements checked.

OCT 16–31 • LEVEL 2

Implement and validate prototype; mentor interaction. Exit: reproducible fault runs, rules, tests and sample explanations.

OCT 31–NOV 4 / NOV 5 ONWARD

Evaluator connection and review preparation. If selected, prepare for regional event and final. Claim progress only when backed by tests.

RISKS & MITIGATIONS

- Overlapping signatures: test ambiguous cases; expose ties and uncertainty.
- Small ring topology: state scope; avoid generalizing to production networks.
- Simulator bias: use held-out runs and report limits.
- Missing telemetry: preserve missing-data markers and allow abstention.
- Multiple faults: acknowledge one-fault-at-a-time assumptions may not hold.
- Unsafe response: no automated remediation.

Methodology: lab first; incremental milestones; unit tests for classifier rules; end-to-end runs after each stage; mentor feedback at reviews; pinned container images, recorded run seed/config hash and one-command demo.

Sources and Scope of Claims

Problem-specific and standards-based sources used to shape the proposal

[1] AIORI Hackathon 3

“PS-035: Evidence-Based Operator Network Fault Localisation,” problem statement supplied for the challenge. Category: Cybersecurity.

[2] RFC 9232 — Network Telemetry Framework

H. Song, F. Qin, P. Martinez-Julia, L. Ciavaglia, and A. Wang. IETF, May 2022. https://www.rfc-editor.org/rfc/rfc9232.html

[3] IETF 126 Hackathon

“AI-Assisted Network Fault Localization and Issue Discovery System,” project description, July 2026. https://wiki.ietf.org/en/meeting/126/hackathon

[4] AIORI Hackathon 3

“Proposed-structure-hackathon.pdf,” official presentation structure supplied for the event.

SCOPE OF CLAIMS

This document presents the proposed design and an early simulator-based starting point. It does not claim production readiness, a completed real network lab, a trained AI model, or independently validated held-out accuracy. Add such claims only after reproducible evidence exists.

Thank You for Your Attention!

THE EDGE INNOVATORS

Evidence-Based Operator Network Fault Localisation

RANK

Prioritise plausible root causes.

EXPLAIN

Show evidence for and against.

ADMIT UNCERTAINTY

Say when evidence is insufficient.

Team code: A3-PS035-TC476

Developed collaboratively by The Edge Innovators

RULE-BASED BASELINE FIRST • NO AUTOMATED REMEDIATION